PROJECT DOCUMENTATION

QUALITY ASSURANCE PLAN

SmartPark – Intelligent Autonomous Parking Ecosystem

PROJECT TYPE	PROJECT TEAM	DOCUMENT TYPE	PROJECT STATUS
IoT, AI, Robotics	**6 Members**	**QA Plan**	**Conceptual Design**

1. PURPOSE AND SCOPE

1.1 Purpose

The purpose of this Quality Assurance Plan is to establish the quality standards, procedures, responsibilities, and evaluation methods required to ensure that the proposed SmartPark system meets its functional, technical, operational, and user-experience requirements.

SmartPark aims to provide an intelligent parking ecosystem that integrates IoT sensors, autonomous robotic valet services, AI-powered parking analytics, and a mobile application. This plan ensures that all system components are designed according to defined quality requirements, work together correctly, and provide a safe, reliable, and efficient parking experience.

1.2 Scope

The Quality Assurance Plan covers the following system components:

1. **Smart Parking Management:** Real-time detection of occupied and available parking spaces.
2. **Intelligent Parking Allocation:** Assigning suitable parking spaces based on availability, location, vehicle requirements, and reservations.
3. **Autonomous Robotic Valet:** Using autonomous robots to transport vehicles between drop-off areas and parking spaces.
4. **Mobile Application:** Providing parking reservations, VIP bookings, vehicle retrieval requests, and parking availability information.
5. **AI Analytics:** Predicting parking demand and recommending less crowded arrival times.
6. **Additional Smart Services:** EV charging integration and automated car washing stations.

The plan also covers documentation quality, system integration, testing procedures, and project deliverables.

> **Scope Limitation:** Since SmartPark is currently a proposed project, quality assurance will focus on design validation, technical feasibility, requirements verification, and proposed testing procedures. Actual performance results will only be reported if a prototype or simulation is implemented and tested.

2. DEFINE QUALITY OBJECTIVES (WHAT AND WHY)

The project will establish measurable quality objectives to ensure that SmartPark meets its intended requirements.

QUALITY OBJECTIVE	WHAT WE WANT TO ACHIEVE	WHY IT MATTERS
Accuracy	Correctly identify occupied and available parking spaces.	Prevents incorrect parking assignments.
Reliability	Maintain consistent system operation.	Ensures uninterrupted parking services.
Safety	Prevent robot collisions and unsafe vehicle transportation.	Protects vehicles, pedestrians, and users.
Efficiency	Reduce the time required to find and retrieve vehicles.	Improves customer experience.

QUALITY OBJECTIVE	WHAT WE WANT TO ACHIEVE	WHY IT MATTERS
Usability	Provide a simple and understandable mobile application.	Makes the system accessible to customers.
Integration	Ensure communication between sensors, robots, applications, and central controller.	Allows all subsystems to operate together.
Prediction Quality	Provide useful parking demand forecasts.	Helps users select less crowded arrival times.
Security	Protect customer information and reservation data.	Maintains privacy and prevents unauthorized access.

Proposed Quality Targets

The following are preliminary acceptance targets for the future prototype or pilot. They are design goals, not verified results:

- **Parking occupancy detection accuracy:** At least 98%.
- **Parking allocation success rate:** At least 99% under normal operating conditions.
- **Mobile application availability:** At least 99% during the defined service period.
- **Reservation conflict rate:** 0%.
- **Successful sensor-to-dashboard updates:** At least 99% in the test environment.
- **Robot obstacle detection:** Successful detection of all defined test obstacles in controlled test scenarios.
- **AI occupancy prediction:** Target MAPE of 15% or lower, where dataset permits calculation.
- **Critical security vulnerabilities:** Zero unresolved critical findings before deployment.

3. ESTABLISH QUALITY CRITERIA

Quality criteria define the conditions that the SmartPark system must satisfy to be considered acceptable.

3.1 Functional Quality Criteria

A. Parking Detection

- Every parking space must have an assigned identification number.
- Sensors must distinguish between occupied and available spaces.
- The dashboard must display the correct parking status. Sensor failures must be detected and reported.

B. Parking Allocation

- The system must assign only available parking spaces; two vehicles must never be assigned the same space.
- VIP reservations must be protected from normal allocation. Vehicle size/requirements must be considered.

C. Autonomous Robotic Valet

- Robots must navigate predefined routes, detect obstacles, respond safely, and include emergency stops.
- Lifting mechanism must support vehicle load; fleet system must prevent conflicting assignments.

D. Mobile Application

- Users can view availability, reserve VIP spaces, request vehicle retrieval, and receive updates/confirmations.

E. AI Analytics

- Process historical/real-time data, evaluate predictions vs actuals, and clearly display predicted periods and limitations.

3.2 Non-Functional Quality Criteria

CRITERIA	REQUIRED QUALITY
Performance	Fast response to parking status updates and user requests.
Reliability	Consistent operation and recovery from failures.
Safety	Compliance with applicable machinery, robotics, and electrical safety requirements.
Scalability	Ability to accommodate additional parking spaces and robots.
Security	Authentication, authorization, and encrypted communication.
Maintainability	Easy troubleshooting and replacement of faulty components.
Usability	Clear interfaces and straightforward navigation.
Compatibility	Support for the selected hardware and software platforms.

3.3 Documentation Quality Criteria

All project documents must: use consistent terminology, follow unified formatting, contain accurate technical data, include diagrams/tables/references, match agreed scope, and avoid unsupported claims.

4. DEVELOP QUALITY METHODS

SmartPark will use several quality assurance and quality control methods:

- **4.1 Design Reviews:** Review architecture, diagrams, component selection, and specs prior to implementation to identify design errors early.
- **4.2 Functional Testing:** Evaluate each subsystem against functional specs (sensor accuracy, space allocation, VIP protection, retrieval routing).
- **4.3 Integration Testing:** Evaluate end-to-end communication across subsystems (Sensor → Microcontroller → Central Controller → Database → Mobile App).
- **4.4 Robotic Safety Testing:** Controlled evaluations of obstacle detection, emergency stopping, navigation accuracy, lifting stability, and loss of communication.
- **4.5 Application Usability Testing:** Evaluate booking ease, availability clarity, retrieval requests, and UI navigation with target users.
- **4.6 AI Model Evaluation:** Evaluate forecasting models against unseen data using MAE, RMSE, and MAPE metrics.
- **4.7 Documentation Reviews:** Conduct peer reviews on all deliverables for completeness, consistency, and accuracy.

5. ASSIGN QUALITY ROLES

Quality assurance responsibilities will be distributed among the six project members.

MEMBER	ROLE	QUALITY RESPONSIBILITIES
Member 1	Project Coordinator	Overall QA supervision, document reviews, final approval coordination.
Member 2	System Architect	Architecture validation, subsystem integration, technical consistency.
Member 3	Robotics Engineer	Robot safety, navigation, lifting mechanism, fleet coordination validation.
Member 4	Software Engineer	Application testing, reservation validation, interface usability.
Member 5	AI Engineer	Data quality, prediction evaluation, analytics validation.

MEMBER	ROLE	QUALITY RESPONSIBILITIES
Member 6	QA & PM Officer	Quality criteria, test documentation, progress monitoring, issue tracking.

Responsibility Structure: Each member owns their subsystem's quality. Member 6 coordinates QA reviews & maintains checklists; Member 2 ensures technical compatibility; Member 1 oversees final deliverables. Critical issues are reported team-wide.

6. MONITOR QUALITY PERFORMANCE

6.1 Key Performance Indicators (KPIs)

KPI	MEASUREMENT METHOD	PROPOSED TARGET
Occupancy detection accuracy	Correct readings / Total readings × 100	≥ 98%
Allocation success rate	Successful allocations / Total allocation requests × 100	≥ 99%
Reservation conflicts	Number of conflicting bookings	0
System availability	Available service time / Scheduled service time × 100	≥ 99%
Application task completion	Successfully completed tasks / Total test tasks × 100	≥ 95%
AI prediction error	MAE or other suitable forecasting metric	Defined post-baseline
Critical safety failures	Number of critical failures during testing	0
Documentation completeness	Completed required sections / Total required sections × 100	100%

6.2 Monitoring Procedures & 6.3 Reporting

The team will conduct daily progress reviews, compare work against quality criteria, record defects, review test results, track corrective actions, and conduct final pre-submission reviews. A Quality Status Report will track tests run/passed/failed, defects by severity, resolution actions, and open concerns.

7. IMPLEMENT QUALITY ACTIONS

7.1 Corrective Action Procedure

1. **Identify:** Record problem & subsystem.
2. **Classify:** Categorize severity (Critical, High, Medium, Low).
3. **Root Cause:** Determine underlying origin.
4. **Develop Solution:** Assign corrective action to responsible member.
5. **Implement:** Update hardware, code, process, or doc.
6. **Retest & Document:** Re-verify fix and record final outcome.

7.2 Example Corrective Actions

IDENTIFIED PROBLEM	CORRECTIVE ACTION
Sensor reports incorrect status.	Recalibrate or replace sensor; repeat detection tests.
Two users reserve same VIP space.	Implement transaction validation and reservation locking.
Robot fails to detect obstacle.	Review sensor config & logic; suspend physical operation until resolved.

IDENTIFIED PROBLEM	CORRECTIVE ACTION
Application displays outdated availability.	Improve data synchronization and communication handling.
AI predictions are inaccurate.	Review training data, features, and model hyper-parameters.
Robot loses communication.	Introduce safe stop procedure and controlled communication recovery.

8. CONTINUOUS IMPROVEMENT

SmartPark follows the **PDCA (Plan–Do–Check–Act)** cycle. Feedback will be gathered from team members, potential customers, technical reviewers, and supervisors. Lessons learned will document problems, solutions, and recommendations.

Future Improvements: SLAM robot navigation, larger AI training datasets, advanced space allocation algorithms, EV charging integration, predictive robot maintenance, accessibility enhancements, and multi-facility support.

> **Conclusion:** This QA Plan establishes a structured foundation for system reliability, safety, and usability. Adhering to these criteria guarantees that the final design provides a robust baseline for prototype development.